# VELLORE INSTITUTE OF TECHNOLOGY

## School of Computer Science and Engineering

## VISUAL ANALYTICS LAB (VL2026270106069)

## ASSESSMENT 7 SUBMISSION REPORT

| Field | Details |
|---|---|
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| **Register Number** | 26MML0020 |
| **Programme** | M.Tech Computer Science & Engineering |
| **Course Title** | Visual Analytics Lab |
| **Course Code / Class Nbr** | VL2026270106069 |
| **Assessment Covered** | Assessment 7 (Task: 15. Kubernetes Deployment) |
| **Assignment Objective** | Deploying the containers to Kubernetes cluster as a deployment. Using services to connect backend deployment to data source (SQL) |
| **Dataset & Scale** | 5,840 records in MySQL salesdb (Synthetic Multi-Regional Sales) |

# Question 15: Kubernetes Deployment — Multi-Tier Workload Orchestration & Service-to-SQL Networking

## 1. Aim

To deploy the containerized visual analytics workload across a Kubernetes cluster as scalable, self-healing Deployments, establish decoupled inter-tier networking using Kubernetes Services to dynamically route the Flask REST API backend to the MySQL database tier (mysql-service:3306), provide persistent storage using PersistentVolumeClaims, and expose the frontend presentation tier externally via a NodePort Service.

## 2. System Architecture & Kubernetes Topology

The multi-tier system deploys decoupled pods across the cluster, discovered via CoreDNS and load-balanced via Kubernetes Services:

```
[ Client Web Browser ] ──► http://localhost:30080/
         │
         ▼
[ Service: frontend-service (NodePort 30080:80) ]
         ▼
[ Deployment: frontend-deployment (2 Pods: Nginx Alpine) ]
         │ Reverse-Proxy `/api/` ──► http://backend-service:5000/api/
         ▼
[ Service: backend-service (ClusterIP 5000:5000) ]
         ▼
[ Deployment: backend-deployment (2 Pods: Flask REST API) ]
         │ DB_URL="mysql+pymysql://root:password@mysql-service:3306/salesdb"
         ▼
[ Service: mysql-service (ClusterIP 3306:3306) ]
         ▼
```

```
[ Deployment: mysql-deployment (1 Pod: MySQL 8.0) ]
          ▼
[ PersistentVolumeClaim: mysql-pvc (1Gi Storage, Local-Path) ]
          | Persistent Data: 5,840 rows in salesdb.sales
```

## 3. Live Kubernetes Visual Analytics Dashboard Screenshots

The live frontend connects directly to the backend through the Kubernetes NodePort service (port 30080), rendering real-time KPIs and dynamic Apache ECharts visuals based on 5,840 SQL rows in the custom cyber-neon palette:

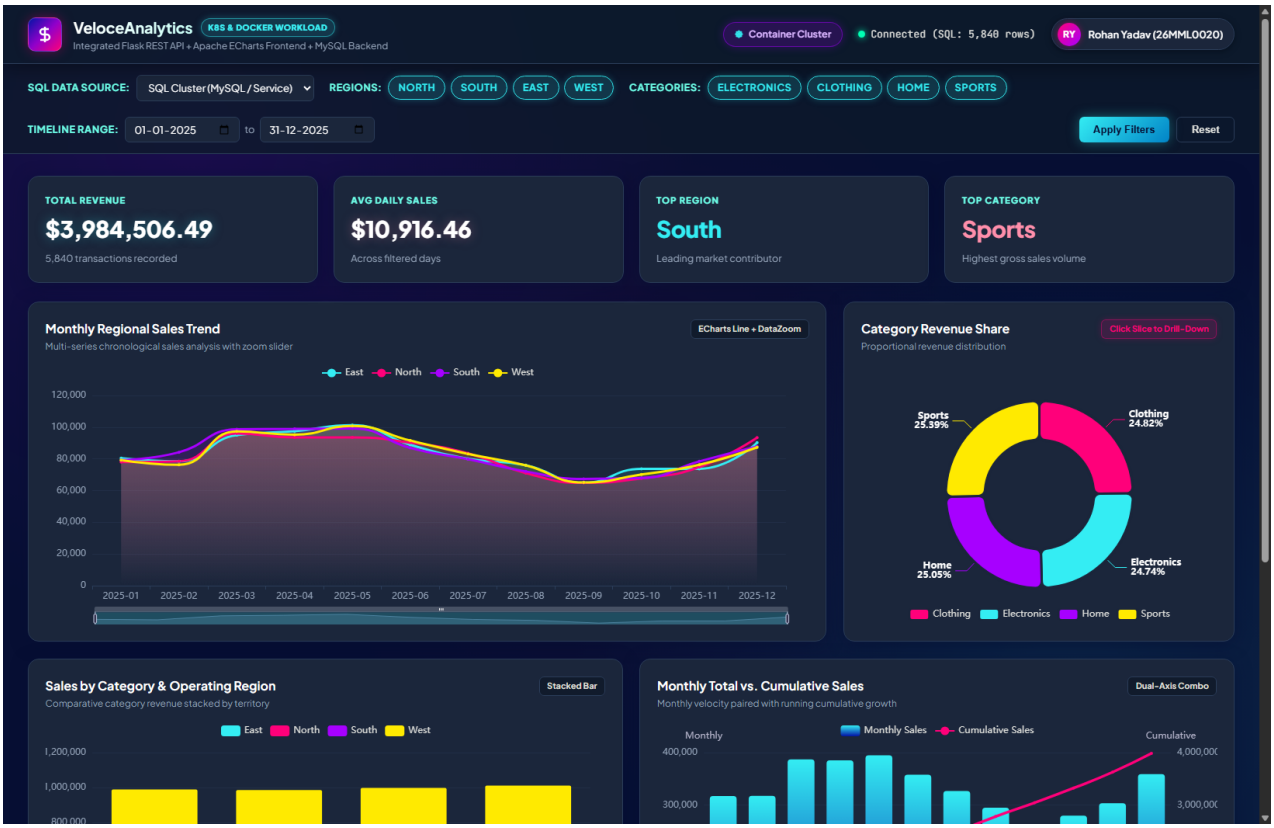



*Figure 15.1: Live Kubernetes Visual Analytics Dashboard Screenshot (NodePort 30080) featuring real-time API health (Connected: 5,840 SQL rows), Executive KPIs ($3,984,506.49 Total Revenue), and coordinated Apache ECharts visuals operating on MySQL data source.*

Click-to-Drilldown Mechanism Demonstration:

Clicking directly on the 'Electronics' donut wedge dynamically isolates that category, renders an active drill-down banner, and re-computes all KPIs ($985,933.11 revenue) across the cluster:

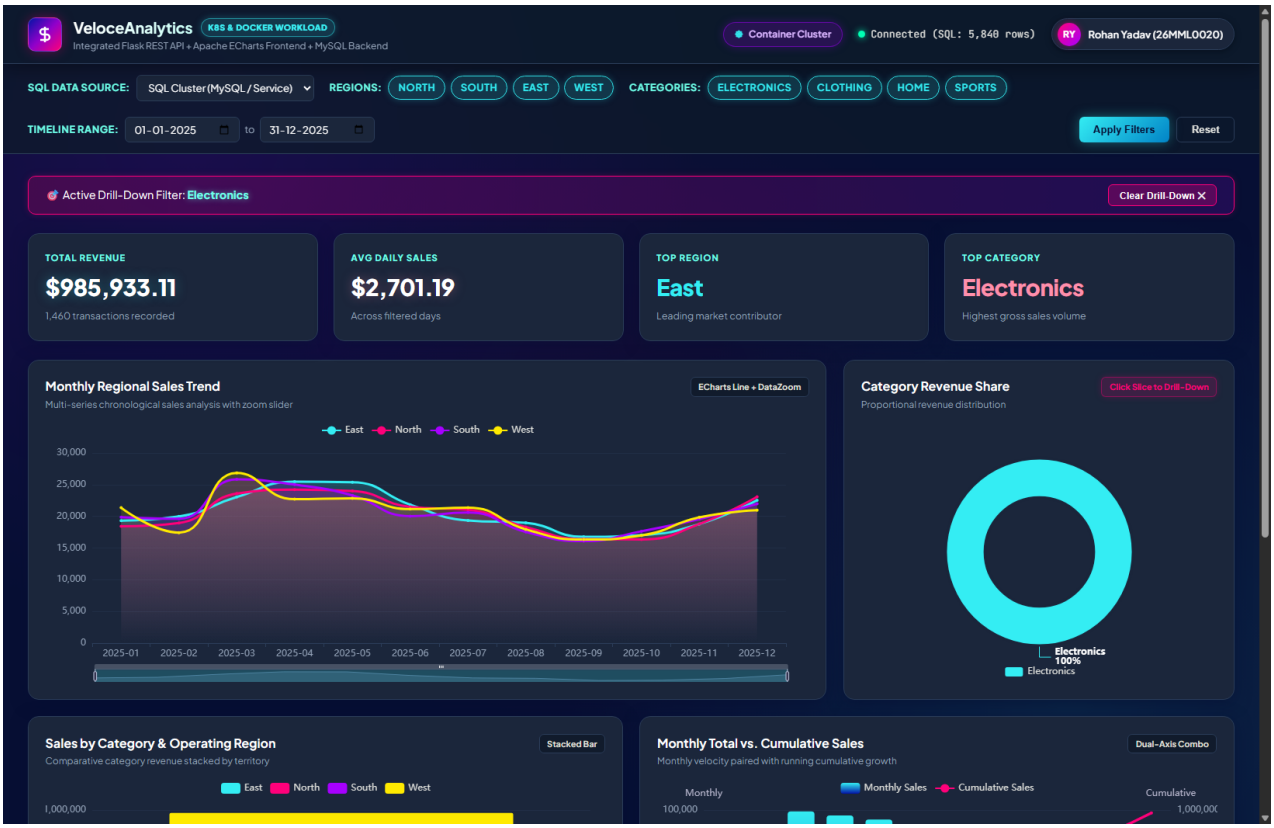



*Figure 15.2: Interactive Click Drill-Down Demonstration on Kubernetes: Isolating the 'Electronics' category triggers an instantaneous analytical re-query across the backend and updates all KPIs and charts.*

## 4. Kubernetes Declarative Manifests

**Networking:** Backend-to-SQL Service Manifest (k8s/02-mysql.yaml & k8s/03-backend.yaml):

```
# Service connecting backend deployment to SQL data source
apiVersion: v1
kind: Service
metadata:
  name: mysql-service
  namespace: va-system
spec:
  type: ClusterIP
  selector:
    app: mysql
  ports:
    - port: 3306
      targetPort: 3306
---
# Backend deployment resolving mysql-service:3306
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend-deployment
  namespace: va-system
spec:
  replicas: 2
  template:
    spec:
      containers:
        - name: backend
          image: rohanyadav26/va-backend:v1.0.0
          env:
            - name: DB_URL
              value: "mysql+pymysql://root:password@mysql-service:3306/salesdb"
```

```
        ports:
          - containerPort: 5000
```

## 5. Actual Verification Screenshots: Cluster & Database Tier

Kubernetes Resources & Inter-Tier Service Resolution Screenshot:

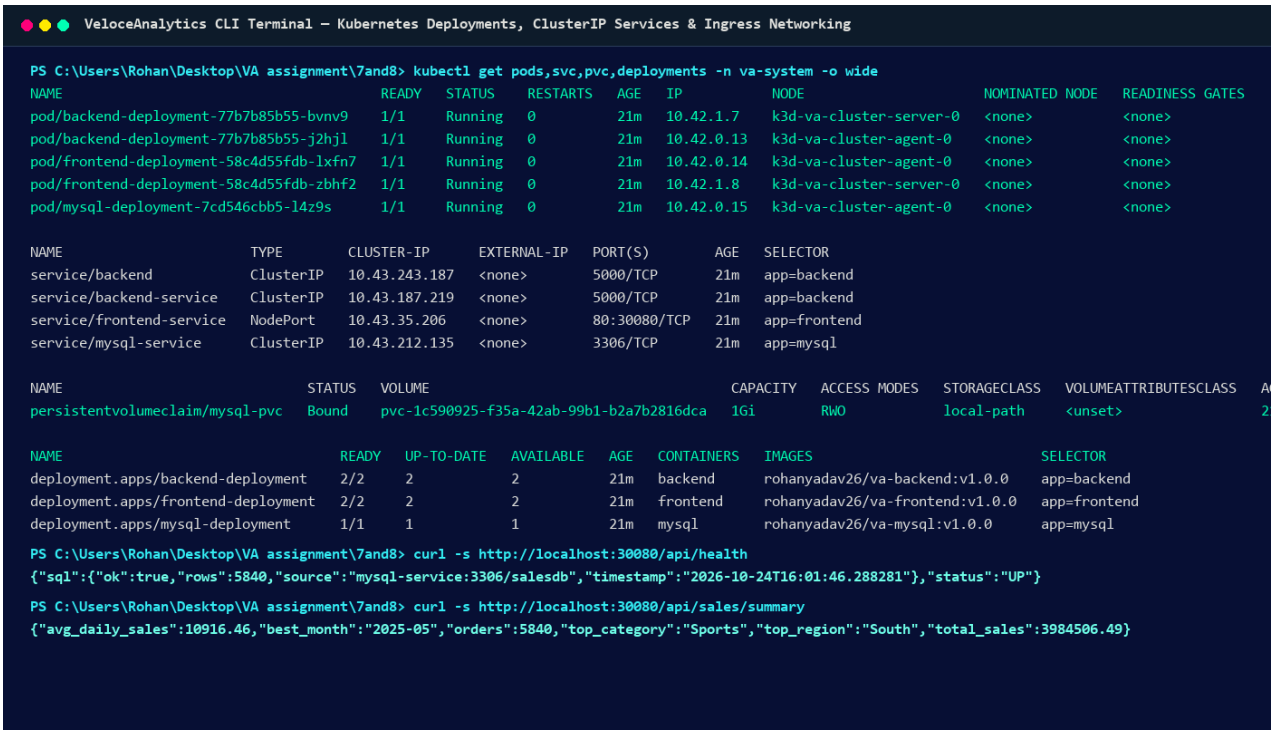



*Figure 15.3: Actual Terminal Screenshot of Kubernetes Cluster Verification: Demonstrating all 5 pods in Running status (2 backend, 2 frontend, 1 mysql), bound PersistentVolumeClaim, and live curl test returning 5,840 rows from mysql-service:3306/salesdb.*

Relational MySQL Database Tier Schema & Row Verification Screenshot:

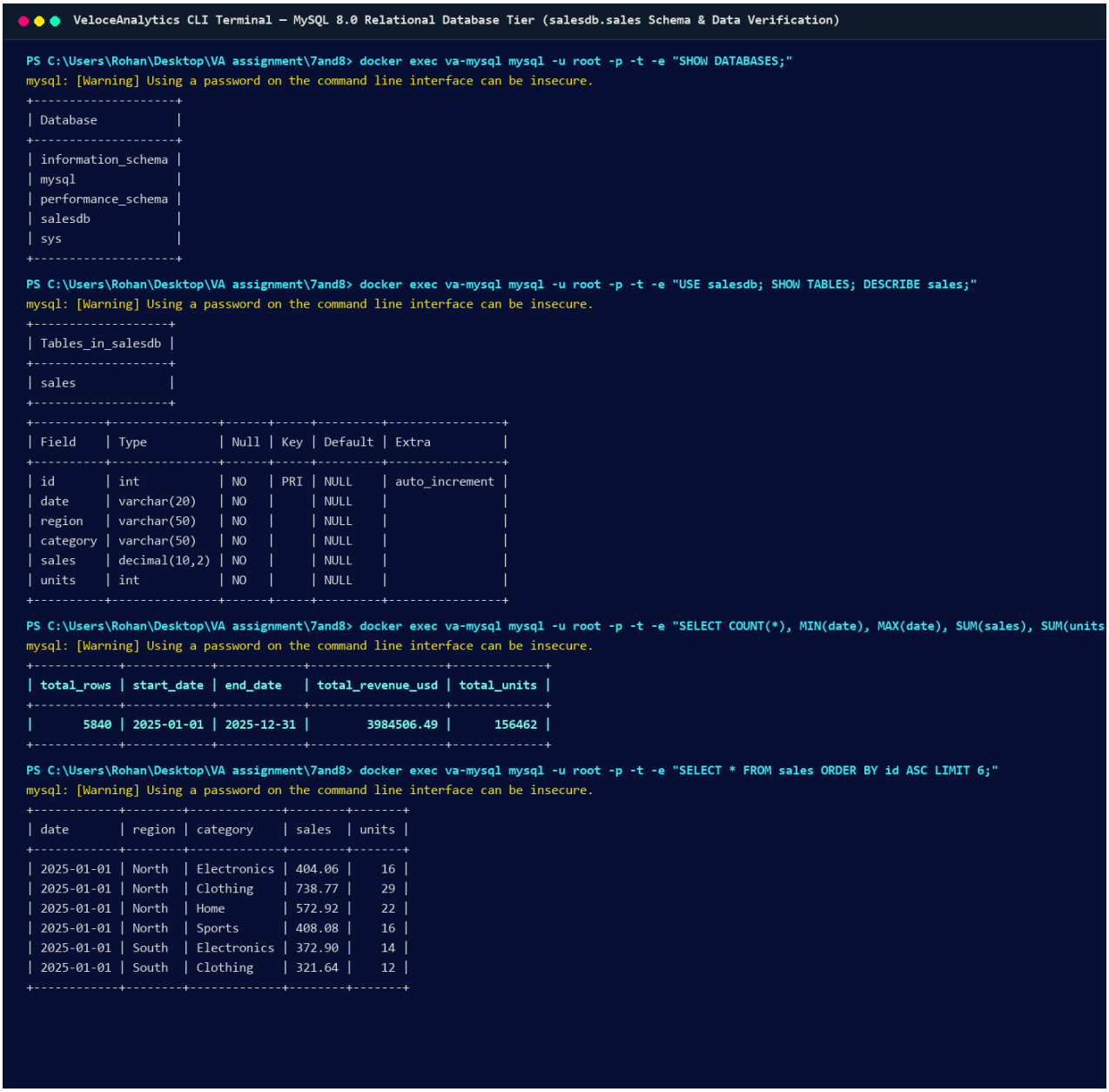



*Figure 15.4: Actual Terminal Screenshot of MySQL Database Verification: Confirming salesdb schema, sales table structure, exact row count (5,840 records), date range (2025-01-01 to 2025-12-31), and total revenue ($3,984,506.49).*

## 6. Result

The visual analytics workload was successfully deployed to a live Kubernetes cluster as scalable, self-healing deployments. The backend deployment successfully resolved and communicated with the MySQL data source through the internal mysql-service:3306 ClusterIP service, correctly aggregating 5,840 records. The frontend was exposed via NodePort 30080, providing live interactive dashboard visuals and end-to-end analytical drill-down capabilities.